

HERMANN PAUWELLS RIVERA

Software Engineer | AI/ML · Full-Stack

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PROFILE

Software engineer focused on AI/ML, specialized in Python (TensorFlow, FastAPI) and full-stack TypeScript (React/Next.js, Node.js), graduating in December 2026 with a Computer Science Engineering degree from Tecnológico de Monterrey. I've developed machine-learning solutions for business cases and full-stack apps for real companies and users (e-commerce, real-time reservations).

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, C++

Machine Learning: TensorFlow, Q-learning, AgentPy

Backend & Data: FastAPI, Flask, PostgreSQL, Supabase, MySQL

Frontend: React, Next.js, Vite

Tools: Git, GitHub, GitHub Actions, Unity

EXPERIENCE

Full-Stack Developer (Freelance) — Juguetería El Arbolito

2026–present

- Developing and implementing an e-commerce and admin platform for a single-branch retail business (catalog, cart, checkout, backend).
- Migrated a 2,395-product catalog from the legacy point-of-sale system (Eleventa) into PostgreSQL/Supabase.
- Implementing Mercado Pago payments and inventory synchronization between online sales and the in-store point of sale.

SELECTED PROJECTS

CrediFiel Challenge (Datathon 2025) Data / ML · team project (Datathon)

2025

- Neural-network model (regression + classification) to optimize Credifiel's direct-debit collections: minimize commission costs or maximize recoveries.
- Real financial dataset of direct-debit collections and efficiency metrics (KPIs); Flask API to serve predictions and a React + Vite frontend.

Multi-agent warehouse simulation (RL) Python · AgentPy · Unity · FastAPI

2024

- Multi-agent Q-learning with conflict resolution, blackboard system and task auctions.
- Animated Unity visualization and a FastAPI server.

Lumina Reservations (WorkHub MTY) Academic team project · Node/Express backend, real-time (SSE) and frontend

2026

- Implemented the reservation backend in Node/Express (services, repositories, controllers, migrations) and real-time occupancy updates over SSE.
- Built the floor map and the reservation flow in React/Vite (dashboard, filters, confirmation) with role controls.

Other projects: folk-park · HeatShield · RainLab Northsky · MAWI · VisualControl (IoT) · Mildred Pierce · GateGenius (HackMTY 2025).

EDUCATION

Computer Science Engineering (ITC) — Tecnológico de Monterrey, Monterrey Campus — 2021–2026 ·

Graduation: December 2026 · Available full-time from January 2027

ACTIVITIES

- SwiftTec Coding Community (2025) — hackathons and innovation challenges.
- Music production and sound design (FL Studio).

Spanish — native · English — C1